India successfully launches Chandrayaan-2 spacecraft

GSLV MkIII-M1 successfully launches Chandrayaan-2 spacecraft (Credit: ISRO)

India's geosynchronous satellite launch vehicle GSLV MkIII-M1 successfully launched 3840kg Chandrayaan-2 spacecraft into Earth's orbit on July 22, 2019. The spacecraft is now revolving round the Earth with a perigee (nearest point to Earth) of 169.7km and an apogee (farthest point to Earth) of 45,475km. The flight marks the first operational flight of GSLV Mk III.

ISRO chairman Dr K. Sivan congratulated the launch vehicle and satellite teams involved in this challenging mission. "Today is a historical day for space science and technology in India. I am extremely happy to announce that GSLV MkIII-M1 successfully injected Chandrayaan-2 into an orbit of 6000km more than the intended orbit and is better.

"Today is the beginning of the historical journey of India towards the moon and to land at a place near south pole to carry out scientific experiments to explore the unexplored," Dr Sivan said.

In the coming days, a series of orbit manoeuvres will be carried out using Chandrayaan-2's onboard propulsion system. This will raise the spacecraft orbit in steps and then place it in lunar transfer trajectory to enable the spacecraft to travel to the vicinity of the moon.

GSLV Mk-III is a three-stage launch vehicle developed by ISRO. The vehicle has two solid strapons, a core liquid booster and a cryogenic upper stage. The vehicle is designed to carry four-ton class of satellites into geosynchronous transfer orbit (GTO), or about ten tons to low earth orbit (LEO).

Chandrayaan-2 is India's second mission to the moon. It comprises a fully indigenous orbiter, lander (Vikram) and rover (Pragyan). Rover Pragyan is housed inside Vikram lander.

The mission objective of Chandrayaan-2 is to develop and demonstrate the key technologies for end-to-end lunar mission capability, including soft landing and roving on the lunar surface. On the science front, this mission aims to further expand our knowledge about the Moon through a detailed study of its topography, mineralogy, surface chemical composition, thermo-physical characteristics and atmosphere, leading to a better understanding of the origin and evolution of the Moon.

Thermal batteries will revolutionise renewable energy production

Curtin University researchers have collaborated with international renewable energy companies, United Sun Systems and ITP Thermal, on the potentially game-changing project, which is being led by Professor Craig Buckley from Curtin's School of Electrical Engineering, Computing and Mathematical Sciences.

Professor Buckley has said that the thermal battery was part of Concentrated Solar Power (CSP) system being developed by United Sun Systems, which requires a battery to store and release energy to enable non-stop solar power generation.

"Storage has long been a stumbling point for renewable energy, but our prototype thermal battery is able to store and, as required, release solar energy without reliance on sunlight, at all times," said Professor Buckley.

"The battery uses a high-temperature metal hydride or metal carbonate as the heat storage medium and a low temperature gas storage vessel for storing the hydrogen or carbon-dioxide.

"At night, and in times of cloud cover, hydrogen or carbon-dioxide is released from the gas storage vessel and absorbed by